Speculative Future for Private Generative LLMs in Dream Interpretation

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As large language models (LLMs) become increasingly embedded in daily life, their potential to mediate introspective practices such as dream interpretation remains underexplored. This research investigates the value of privately deployed, fine-tuned LLMs as personalized tools for dream reflection. Situated in a speculative near-future scenario, the study compares the user experience, perceived privacy, and long-term viability of private versus public cloud-based LLMs in the context of dream analysis. Drawing on a mixed-methods approach, researcher conducted structured dream reflection sessions with 15 participants using both ChatGPT and a locally deployed, fine-tuned LLM. Quantitative and qualitative data reveal that public LLMs prompt self-censorship and emotional withholding, aligning with contextual integrity theory. These findings highlight the psychological and ethical stakes of AI-mediated introspection and offer design implications for culturally adaptive, privacy-preserving LLM interfaces in personal well-being practices.

CCS CONCEPTS • Private LLM • Dream Interpretation • Human-AI Interaction

Additional Keywords and Phrases: Human-Computer Interaction, Dream Reflection, User Experience

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1 INTRODUCTION

As a universal and profound phenomenon of human psychological activities, dreams have long been intertwined with philosophy, literature, art, and science.[1] As a cross-cultural experience, dreams have inspired multiple interpretations of different civilizations, reflecting different worldviews and cosmic views. From ancient philosophers to contemporary scholars, human interest in dreams has never diminished, and their symbolic meanings, psychological functions, and aesthetic potentials have been continuously explored and reshaped.[2] Although with the development of biomonitoring and neuroimaging technologies, researchers have been able to observe brain activity during sleep more accurately, there are still many unknowns about the specific role of dreams in cognitive regulation, emotional processing, and mental health.

Dreams are physiological and psychological states of consciousness that occur during sleep, usually characterized by vivid sensory, motor, emotional, and cognitive experiences. They most often occur during rapid eye movement (REM) sleep. Dream interpretation—also known as dream analysis—refers to the process of assigning meaning to dreams [3]. Although dreams are fleeting and often fragmented, they are a unique window into the subconscious. For people, recording dreams is not only a way to express emotions or reflect on themselves, but also a means to understand underlying psychological mechanisms.[4]However, the nature of dreams—ephemeral, fragmented, and highly personal—makes them difficult to recall and interpret, especially in the absence of systematic support. Traditional dream recording methods, such as handwritten diaries or digital note-taking apps[5], [6], while helpful in preserving content, often lack structured support and deep interpretation capabilities.

In recent years, the rapid development of large language models (LLMs) has provided a new path for processing and interpreting dream content. With their natural language understanding and generation capabilities, LLMs can help users identify recurring imagery in dreams, analyze emotional clues, and even generate guided questions to help users reflect more deeply on their dream experiences. However, this process also faces a key challenge—dream reports often involve extremely private information, such as traumatic experiences, identity anxiety, and sexual fantasies. Users may find it difficult to trust such data to be uploaded to cloud platforms for processing.

In this context, private LLMs deployed on local devices have gradually attracted attention. [7]Such models can run model inference without relying on network connectivity while ensuring that users have full control over their data. Private LLM allows for reflective interactions without uploading sensitive content to external servers, creating a safer space for emotional vulnerability and creative exploration. The modifiability and controllability of private models also make private LLM a more personalized system. Therefore, it can serve as an intelligent companion to assist users in personal information management, provide language support when users review their dream experiences, and understand their dream experiences. It can identify recurring symbols in conversations, track different emotional themes in dreams, and even generate guided questions to promote deeper reflection and generate personal dream informatics.

Even in the context of increasingly mature technology, the subjectivity and high symbolism of dreams still pose challenges to automated interpretation. Dream texts are difficult to standardize, and there are significant differences in the interpretation of the same image in different cultures. In addition, how automated systems can avoid "over-rationalizing" dreams or misinterpreting their subjective meanings while providing structured support is still a question that needs to be explored in depth. Researchers must strike a balance between system intelligence and user autonomy to build interactive experiences that truly support reflection and growth.

1.1 Research Goal and Research Question

This study seeks to explore the potential of locally deployed, fine-tuned large language models (private LLMs) as personal tools for dream interpretation. By situating the investigation within a speculative future context, the research aims to understand how such models may support more private, personalized, and meaningful engagement with dream content. Specifically, the study compares private LLMs with publicly available, cloud-based LLMs in terms of user experience, perceived privacy, and long-term adoption potential.

To guide this inquiry, the main research questions are proposed:

In the speculative future context, how does private LLM support user interaction for personalized dream interpretation?

Sub-questions:

- RQ1: What are the limitations of current public LLMs in dream interpretation?

- RQ2: How do private tailored (fine-tuned) LLMs improve the experience and outcomes of dream interpretation?
- RQ3: Do users see value in adopting private LLMs for dream interpretation as a long-term, personal practice?

1.2 Contribution

The research contributes to a new and underexplored intersection between AI-driven cognitive analysis and ethical AI design. The findings will provide practical implications for user-centered AI applications, especially in sensitive areas such as mental health and self-reflection tools. AI-assisted construction of personal information systems. By systematically exploring the above issues, this study identifies the technical and ethical advantages of locally deployed private LLM in dream analysis applications and identifies key challenges in its design and practical use. Particular attention is paid to the model's potential in promoting user self-reflection, building a safe expression space, and supporting emotion regulation. At the same time, it is also hoped to evaluate how users make trade-offs between functionality, privacy protection, and interpretability when using private LLM and cloud LLM. This exploration will provide a theoretical basis and design suggestions for the future development of personalized and controllable dream analysis systems, and promote the practical application of private semantic interaction in mental health and daily self-management.

1.3 Delimitation

This study does not aim to optimise language-model algorithms, rather, it investigates how a privately deployed LLM can be designed and experienced in the highly sensitive context of dream interpretation. Consequently, the underlying model—LLaMA 3.1 (8 B parameters)—received only light domain adaptation, no bespoke dream corpus or intensive fine-tuning was undertaken. Moreover, the model experiment relied on a single architecture, without systematically benchmarking alternative open-source models (e.g., Mistral, Gemma). This constrains the generalisability of the findings to other model families and parameter scales.

Dream interpretation itself is inherently subjective and symbolically polysemous. Hence, all AI feedback in the study was framed as an aid to personal reflection, not as clinical or psychotherapeutic advice. This study did not evaluate the factual or psychological accuracy of model interpretations; instead, research focus lay on Human‑AI Interaction (HAI) dimensions—trust calibration, perceived control, and emotional resonance.[8]

Although a two-model comparison was conducted, the private LLM remains functionally less mature than state-of-the-art cloud models. The comparative study therefore serves exploratory purposes: to identify where small, local models might evolve and differentiate themselves within a cloud-LLM-dominated landscape, rather than to claim parity in raw performance.

2 RELATED WORK AND THEORY

2.1 Theoretical Foundations of Dream Interpretation

Ancient cultural practices interpreted dreams in a symbolic and spiritual way. [9] [10] [11] [12]Dream interpretation has long been a topic of interest in the fields of psychology, neuroscience, and artificial intelligence(AI). However, before 1900, European intellectual history generally believed that dreams were meaningless. Before that, people generally believed that dreams were communications between God and dreamers. Against this background, the emergence of ***"The Interpretation of Dreams"*** restored the meaning of dreams, and described the origin of dreams and the function of dreams on dreamers from a scientific perspective, and formed a series of coherent scientific theories such as psychoanalysis.

Freud's psychoanalytic theory believes that dreams are a window to the subconscious, revealing repressed thoughts and desires. [3] He believes that nighttime dreams often recreate past events, but place them in a different context. Frued believes that what happens during the day will activate people's subconscious desires and conflicts, and the content of dreams will often unfold based on this. In this sense, although dreams distort the content, they reveal the theme of the subconscious. Freud also proposed that dreams can be regarded as communication about subconscious content in the therapeutic relationship. In this regard, Freud laid the foundation for later theories related to dream therapy. [13] But his dream theory has been controversial in the psychoanalytic community since its inception, including his student Carl Jung. In contrast to Freud, Jung described dreams as images that spontaneously depict actual situations in the subconscious in symbolic form. Jung believed that the subconscious is not only composed of repressed things, but also contains collective components that play a constructive role in personality development. [14] Later, Bion's theory believed that dreams and subconscious thinking have creative abilities, can resolve conflicts, generate new ideas and promote psychological growth. [15]Similarly, Donald also recognized that *"Dreams are creative processes which generates meaning that can be deployed to life and relationships in the outside world."* [16]

Among them, the latest theory is the Control-Mastery theory (CMT) proposed by Gazzillo.

> *"A dream is an unconscious attempt to find a solution to an emotionally relevant concern. In dreams people think about their main concerns, particularly those concerns they have been unable to solve by conscious thought alone, and try to develop and test plans ... for dealing with them. ... From this perspective, dreams may be viewed as simple but important messages that dreamers send themselves. "*[17]

Therefore, regarding the role and significance of dream interpretation, there is convincing evidence that dreams are fundamentally meaningful and not meaningless fragments of garbage in the brain. Many studies have shown that dreams are not only closely related to the dreamer's waking life, but also to the emotionally significant and inner conflicts in the dreamer's memory. Dreams are not only related to experiences in waking life, but also process this information in a beneficial way. Windt summarized the connection between dreams and memories as follows: emotionally meaningful or repressed experiences will be activated from short-term memory during dreams, and problems will be redone or the possibility of solving problems will be provided. The redone content may be stored back in long-term memory and will not cause psychological burden in real life. Failure to redo may cause waking up and become a nightmare [18]. Dreaming is functional and can regulate people's psychological functions.

2.2 AI and Computational Dream Interpretation

With the development of AI technology, researchers have begun to apply Natural Language Processing (NLP) to the analysis of dream texts. Traditional dream interpretation relies on manual annotations and expert interpretation, which is inefficient and highly subjective. Modern language models, in recent years, language models based on deep learning have made significant progress in text understanding and generation, providing new tools for dream interpretation. For example, Cortal proposed a sequence-to-sequence (Seq2Seq) language model for automatically identifying characters and emotions from dream narratives, realizing structured analysis of dream content [19] . In addition, the DreamNet framework developed by Panchagnula et al. [20], combined with the Transformer architecture and multimodal attention mechanism, can extract semantic themes and emotional states from dream texts, achieving an accuracy of 92.1% in plain text mode, showing potential in mental health diagnosis and personalized treatment.

In order to improve the performance of the model on specific tasks, the researchers used fine-tuning technology. For example, the Dream 7B model developed by the NLP team of the University of Hong Kong significantly improved the performance of the model in dream interpretation tasks by fine-tuning it on 1.8 million conversation data [21]. In addition,

Hua and Yu used a fine-tuned LLaMA 3.1-8B model to analyze the dream content shared by users on the Reddit platform, achieving accurate classification of dream emotions, providing data support for studying the impact of the epidemic on the public psychology [22].

An important research area is the application of electroencephalogram (EEG)-based models, as demonstrated by the Dream Visualization Center of Tsinghua University. Researchers at the center use EEG signals collected during sleep to generate visual representations of dreams, providing a unique way to externalize subconscious imagery. Such research aims to transform implicit cognitive states into explicit visualizations. [23] Another key development is the application of artificial intelligence in dream content classification and emotion analysis. For example, some studies use artificial intelligence to encode dreams by extracting themes and emotions, helping psychologists organize and summarize dream reports. [24] [25] An important study by Laureano compared human dream interpretation with AI-generated interpretations, highlighting differences in emotional depth and symbolic meaning extraction. [26] In addition, Juncker explored the potential of ChatGPT in dream interpretation, iterating the AI model to enhance its ability to analyze dream narratives. His research has sparked discussions about the intersection of AI-driven cognitive analysis and traditional psychological methods, and raised questions about the role of machine-generated interpretations in professional practice. [27]

2.3 Cloud LLM & Private LLM

In the field of artificial intelligence, LLMs are mainly deployed in two ways: Cloud LLM and Private LLM. These two deployment methods have their own advantages and applicable scenarios. Especially when dealing with sensitive data such as dream interpretation, it is particularly important to choose the right deployment method.

Cloud LLM is a large language model deployed on the cloud service provider platform. Common cloud servers include OpenAI, Google, etc. Users access these models through the Internet and only need API interfaces to use them. There is no technical threshold. Cloud service providers have powerful computing resources and can adapt to tasks of different scales. Cloud LLM (such as GPT-4, Gemini, Deepseek) relies on powerful computing resources and massive data training to have excellent text understanding and generation capabilities. In dream interpretation, these models can automatically identify characters, emotions and themes in large sections of dream descriptions and realize structured analysis. However, cloud LLM has challenges in data privacy. User data needs to be uploaded to an external server and may be further trained by the user model, which may leak sensitive information. In addition, the memory mechanism of cloud models is usually opaque, and it is difficult for users to control or delete stored data. This may cause users to have anxiety and trust issues when dealing with dream content involving personal privacy.

Private LLM refers to large language models deployed on the user's own hardware or private cloud environment, such as open source models such as LLaMA and Mistral. The user has full control over the model, including data, model parameters, and operating environment. This deployment method can fully control the storage and processing of data and avoid uploading sensitive information to the cloud, thereby enhancing data privacy and security. Private LLM provides a more controllable solution. Locally deployed models usually store data in user-defined directories, providing the possibility of full data ownership, selective memory, and user-controlled deletion. Emerging architectures also allow for modular "episodic memory" structures, which, if explicitly enabled, can retain context between sessions, thereby simulating diary records without compromising privacy. These mechanisms meet users' desire for continuity in reflective practice while respecting ethical boundaries in terms of informed consent and psychological safety. [28], [29]

The development of private LLM enables more individuals to deploy personalized AI models, and models of different sizes can meet different functional requirements. The rise of open platforms and open source models from different companies provides users with more opportunities to try and customize AI solutions. [30] Lukas et al. outline various

strategies for deploying private LLMs, highlighting options for users with different computing resources and privacy requirements. [7] These advances pave the way for safer and more personalized AI applications, such as dream interpretation. Hu et al. proposed a fine-tuning method called LoRA fine-tuning, which is a low-cost method that requires only minor code adjustments to train LLMs. [31] These training methods make it easier for individuals to DIY their own LLMs and use private AI as a personal tool to meet specific needs.

However, deploying private LLMs also faces certain technical and resource challenges. Users need to have certain computing resources, such as high-performance GPUs and technical capabilities, to ensure the effective operation and maintenance of the model. In addition, although private LLMs have models of different sizes and different data volumes to meet different functional requirements, their performance may not be as good as cloud models, especially when processing large-scale data or complex tasks.

2.4 AI Anxiety and Trust

The rapid popularity of generative AI tools has triggered growing public concerns, which are often referred to as AI anxiety - a psychological reaction that includes fear, suspicion or discomfort about the capabilities and social roles of AI. [32] There has been no serious privacy breach news about AI, but this does not affect people's anxiety and concerns about AI. Governments of various countries have also set different degrees of restrictions on AI.[33], [34] Frenkenberg summarized that there are two dimensions of AI anxiety: anticipatory anxiety, driven by fears of future disruption, and annihilation anxiety, reflecting existential concerns about human identity and autonomy. Regarding the use of AI for dream interpretation, this anxiety is often exacerbated due to the sensitivity and privacy of the data involved. Anxiety about AI has a U-shaped relationship with usage, where moderate engagement reduces anxiety, and high or low levels increase it. This study focuses on trust in AI during long-term use, so users will use it more frequently and need a higher sense of control and understanding of the system. [35] Zahra analyzed 90 studies and found that there is a statistically significant positive correlation between the explainability of AI systems and user trust. Although explainability helps to build user trust, it is not the only or most important factor. [36]

Therefore, when users perceive AI tools as transparent, empathetic, and under their control, they are more likely to trust these tools. When AI systems analyze deeply personal content such as dreams, trust must be earned not only through accuracy, but also through respectful engagement and explainability. Some studies have emphasized the importance of explainability and emotional fit in building user trust, especially in systems involving mental health, counseling, or introspection. For example, AI empathy, accountability, and explainability in depression treatment can increase patients' privacy concerns and trust levels. [37]Using Private LLM can give users greater transparency and control over their data and interpretation process, thereby alleviating some AI anxiety. The increase in user trust in AI systems is closely related to their perception of transparency and controllability of AI decision-making processes. When dealing with sensitive personal information, users are more inclined to trust AI systems that provide clear explanations and allow users to control the use of data. [38]

2.5 Human-AI Interaction

Human-AI Interaction (HAI) is an important branch of human-computer interaction (HCI), focusing on the collaborative mechanism between humans and artificial intelligence systems with certain autonomy and reasoning capabilities. As AI systems transform from passive tools to *"actors"* or *"collaborators"* [39], the theoretical framework of HAI has undergone three key transformations: role reconstruction, trust construction, and redefinition of control mechanisms. Lee and See [40] proposed a "trust calibration" model, emphasizing that users' trust in AI systems should be dynamically adjusted based on

their capabilities and the predictability of their behavior. Shneiderman [41]pointed out that AI should enhance rather than replace human decision-making, and proposed the concept of "adaptive autonomy", that is, the system can act autonomously when necessary while allowing humans to retain ultimate control.

Based on these theories, HAI research has further developed a series of core design principles, including transparency, controllability, and contextual integrity[8], [42]. Especially in highly sensitive scenarios, these principles are of greater importance: the system needs to reveal its decision logic, respect the user's right to edit content, and ensure that data processing complies with privacy expectations in the context. The private LLM model design, dream input format template, and interactive output strategy in this study are all deeply influenced by HAI theory, aiming to establish a trustworthy human-computer collaborative relationship and balance the tension between cognitive support and emotional vulnerability protection.

3 METHOD

3.1 Research Approach

This study adopts a **usability lab study** and a **comparative study** framework to systematically investigate the differences in user experience between a **cloud-based LLM** (ChatGPT) and a **locally deployed private LLM** in the context of dream interpretation tasks. A mixed-methods approach was employed, combining **quantitative questionnaires** and **qualitative semi-structured interviews** to achieve a well-rounded evaluation.

3.1.1 Usability lab studies

The usability lab study method[43], which enables the capture of detailed user interactions in a controlled environment, is essential for evaluating differences in user experience between cloud-based and on-premises LLMs for sensitive tasks such as dream interpretation. This method enables direct observation of user behavior, allowing researchers to capture not only user actions, but also their reactions, hesitations, or confusion. Given the emotional nuances and personal reflectiveness of dream interpretation, live reactions and engagement patterns are essential for understanding user experience beyond surface satisfaction.

In this experiment, usability lab studies were incorporated into the research design in the following ways: Both LLMs were deployed on the researcher’s own machine, ensuring a stable and consistent interaction environment for all participants. This minimized differences due to network latency, inconsistent interfaces, or device differences. All participants were asked to complete the same dream interpretation prompts with each model, allowing for direct comparison of model performance and user perception at the task level. During the experiment, Researcher observed participants' interactions in real time and recorded behavioral cues. Researcher then triangulated these observational data with questionnaire results and qualitative interview results to deepen the analysis. Participants interacted with the two models in sequence, and the order of interactions (A→B or B→A) was randomized to balance potential sequence effects. This allowed each participant to serve as their own control, thereby enhancing the internal validity of the comparison.

3.1.2 Comparative study

This study focuses on comparing the similarities and differences between private LLM and mature cloud LLM (ChatGPT) in dream interpretation tasks, and explores the performance and potential of small models in the field of dream interpretation. Comparative study can directly reveal the differences between the two models in terms of parsing ability, user experience, and privacy protection. [44], [45]Through comparative study, Researcher can gain an in-depth

understanding of the differences in user preferences and experience when using the two models, especially in dream interpretation, a sensitive task involving privacy and emotion. This method can provide direct user feedback for the optimization of small models. Comparative study can also clearly point out the positioning and development direction of small models in the market dominated by LLMs, providing a clear path for future research and development.

3.2 Materials

3.2.1 Model Configurations

The two models used in the comparative study need to be deployed in advance on the computer used in the experiment. The specific properties of the models are shown in Table 1. The two models are:

Model A: a general cloud LLM model without fine-tuning (ChatGPT 3.5).

Model B: a fine-tuned private LLM model trained specifically on dream-related prompts and interactions.

Table 1: Features of two model

Feature	Model A – Cloud LLM	Model B – Private LLM
Base architecture	GPT-3.5 (OpenAI)[46]	LLaMA-3.1-8B
Deployment	API (remote server)	Ollama local runtime
Fine-tuning	None	Supervised LoRA fine-tune (Unsloth)[47]
UI shell	Gradio[48], ChatGPT-style	Identical Gradio UI
Typical latency	1 – 3 s	20 – 30 s

3.2.2 Model B：The process of building and fine-tuning a private LLM

In order to better fit the emotion-intensive and culturally relevant task of dream analysis, this study built and deployed a private LLM (Model B) based on the HAI theory. Its basic model is LLaMA 3.1 8B, and it is run using the local deployment framework Ollama. [49]Compared with directly calling the open source basic model, this study has carried out multiple rounds of customized optimization of the model around the specific needs of dream analysis, including corpus preparation, prompt template design and supervised fine-tuning, to ensure that it has a higher contextual grasp and symbolic association ability when understanding complex dream content. The specific model technology process is shown in Figure 1:

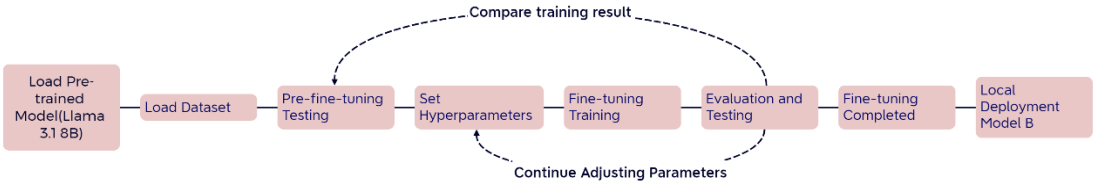


Figure 1: Technical implementation process

3.2.2.1 User needs research and model design alignment

In order to clarify users' expectations and needs for AI assistants in dream analysis, this study conducted a targeted user survey at the beginning of model design. Researcher interviewed 5 potential users with an average age of 24 who had dream analysis experience. Among them, 3 had the habit of recording dreams on a daily basis and had a certain background in dream interpretation. They would usually communicate with professional dream interpreters; the other 2 participants would actively seek dream interpretation services when they encountered special dreams. Each interview lasted about 30 minutes and was conducted in a semi-structured way. It revolved around the real experience of users interacting with dream interpreters, focusing on the dimensions and information needs they paid attention to in the dream interpretation process (Appendix A).

The model fine-tuning process took user needs as the starting point, integrated the results of literature research and initial interview analysis, interview questions and clarified three core functions: keyword extraction and symbol recognition, multi-dimensional analysis capabilities, and support for the complete generation and memory reconstruction of dreams.
In combination with the above user needs, Model B (private LLM) incorporates the following three design points when it is constructed:

1. Keyword recognition and symbol extraction: Through named entity recognition (NER) and symbol density analysis and other technical means, high-frequency images or core symbols in dreams, such as "water" and "falling from a high altitude", are extracted as analysis entry points;
2. Multi-dimensional interpretation module: Fine-tune the embedded psychology (such as Jung's archetype theory) and cross-cultural symbols (such as Chinese and Western dream dictionaries) in the corpus to support the generation of interpretation content from multiple perspectives;
3. Structured output design and user interaction flexibility: Design a three-paragraph answer template and maintain an interface style consistent with mainstream AI chat products, so that users can participate in customized interactions by modifying questions.

3.2.2.2 Model deployment and fine-tuning process

In order to meet users' needs for dream keyword extraction, multi-dimensional analysis and structured narration, Model B adopts a private deployment solution based on the open source LLM architecture, combined with specific corpus for fine-tuning and prompt engineering.

Model selection and local deployment：

This study chose a small-scale language model with open weights and support for local deployment, taking into account both performance and privacy requirements. The model is hosted on a local server and provides controllable chat services through Docker and API interfaces to ensure that the data does not leave the local server and meets the user's confidentiality requirements for dream content.

Corpus preparation and data annotation

The fine-tuning corpus mainly comes from two parts:

1. Structured dream analysis samples: Referring to the text style of the dream interpreter mentioned in the interview, combined with the dream dictionary on the Kaggle platform, researcher constructed 200 sets of training samples

consisting of "dream description + interpretation text + keywords + multi-angle interpretation", covering common images (such as water, flying, chasing) and their cultural symbolic meanings;[50]

2. Interactive dialogue corpus: Collect dream reports on the open source dream database dreambank, simulate the question-and-answer dialogue between users and dream interpreters, annotate the user's question type (such as emotion-oriented, symbol-oriented, causal relationship-oriented) and the corresponding answer strategy, to enhance the coherence and pertinence of the model in multi-round dialogue scenarios.[51]

These corpora were manually organized and structured into JSON format, with clear input fields (user_prompt) and target output fields (dream_interpretation, keywords, explanation_dimensions) for supervised fine-tuning.

Model fine-tuning and parameter optimization

Researcher used the LoRA (Low-Rank Adaptation) for lightweight fine-tuning, and enhanced its performance in the vertical task of "dream interpretation" while retaining the model's original language understanding ability. During the fine-tuning process, the model's performance in the following capabilities was optimized:[31], [52], [53]

1. Keyword recognition and context extraction: The model was trained to accurately identify symbolic images and psychological events in dreams;
2. Multi-dimensional association generation: By constructing labeled interpretation dimensions (such as psychological perspectives and cultural metaphors), the model was trained to cover multiple analysis paths when generating content;
3. Answer style control: Introducing the common tone of dream interpreters (such as encouragement, inquiry, and symbolic association) as soft prompts to guide the model to output texts with more humanistic care and inspiration.

The training uses the GGUF 8-bit model loading method to reduce resource consumption. [54]After training, the model is connected to the chat interface through the API. The specific deployment steps are shown in Figure 2:

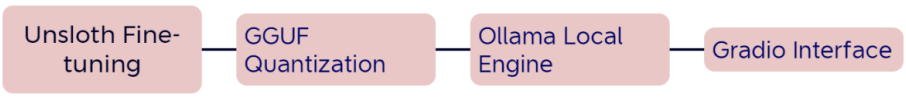


Figure 2 :Model deployment phase

Interactive interface integration and response mechanism design

The final model is deployed to the Web front-end interactive prototype. Users can upload dream text or keywords in the interface. After the model is processed by the Figure 3 framework, it returns the parsed content in a three-segment structure, including:

1. Keyword summary segment: List the core images in the dream;
2. Multi-dimensional interpretation segment: Interpret from the perspectives of psychological symbols, cultural symbols, situational hints, etc.;
3. Memory construction segment: Help users review the dream plot in the second person, enhance self-understanding and emotional organization.

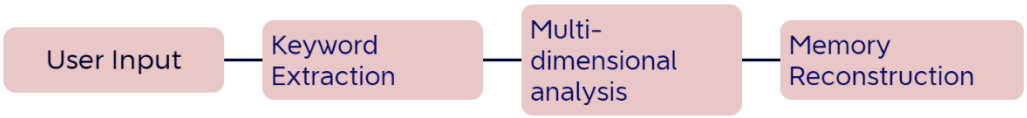


Figure 3 :Model runtime operation

3.2.3 Questionaire(Appendix A.2)

The Likert scale was chosen as the primary measurement tool for the post-interaction questionnaire because it has been shown to effectively capture users' subjective attitudes and experiences in a structured and quantifiable way. It allows participants to express their level of agreement or disagreement with a series of statements, which is particularly useful for assessing underlying concepts such as trust, usability, emotional resonance, and perceived privacy. The questionnaire was a self-designed questionnaire based on the research questions and core comparison dimensions. Since both models provide the same core dream interpretation function, the questionnaire items were kept consistent to ensure consistency and comparability across conditions.

The questionnaire was scored using a Likert scale (1-5), representing: 1 – Strongly Disagree, 2 – Disagree, 3 – Neutral 4 – Agree, 5 – Strongly Agree

Compared to binary ratings, this design is able to measure user perceptions more finely and capture subtle differences in experience.

The questionnaire contains 15 items covering the following four dimensions:

Usability (Q1, Q2, Q8)

Trust and Privacy (Q6, Q11, Q12, Q13)

User Experience and Perceived Meaning (Q3, Q5, Q9, Q10, Q14)

Willingness to Repeat and Willingness to Use in the Future (Q4, Q7, Q15)

This structure ensures a balanced and comprehensive evaluation of each model from multiple perspectives.

3.2.4 Semi-structured interview(Appendix A.3)

To complement the questionnaire and gain deeper qualitative insights, a semi-structured interviews was conducted with participants after they completed interactions with both models. The interviews were designed to explore users' perceptions, preferences, and concerns in more depth, especially in relation to the core research questions.[55], [56]

The interview questions focused on three key areas:

1. Users' understanding and attitudes toward dream interpretation – to contextualize how users perceive the role and meaning of dream interpretation in their personal lives.
2. Comparison between the two models – to gather subjective feedback on which model participants preferred and why, focusing on factors such as interpretation quality, personalization, and engagement.
3. Trust and privacy in AI interaction – to explore users' willingness to share personal content with AI, their concerns about data security, and how these concerns may influence their behavior during interaction.

The semi-structured format allowed for flexible, participant-led discussion, while still ensuring that each interview addressed the core themes relevant to the study.

3.3 Paricipants

Participants in this study were aged between 18 and 30 years old (mean age 22 years old). Participants in this age group were selected because they were familiar with AI and LLM technology and had a strong interest in AI-related applications. To ensure that the study covered a wide range of perspectives, participants from different academic backgrounds were selected. About 80% of the participants were frequent users of generative AI, and many of them said that they often used tools such as ChatGPT, DeepSeek and Kimi to complete academic tasks and daily problem solving. Participants were recruited without restrictions on knowledge of dream interpretation.

A total of 15 participants participated in this study. Table 2 summarizes each participant's academic background, LLM-related experience, and cultural familiarity related to dreams.

Table 2: Participants list

Participants	Age	Major	LLM engagement level	Private LLM Experience	Dream Interpretation Experience
P1	25	Human-Computer Interaction	Moderate	Not familar	Familiar
P2	19	Pharmaceutical Engineering	Low	Not familar	Not familar
P3	18	Pharmaceutical Engineering	Low	Not familar	Not familar
P4	18	Chinese materia medica	Low	Not familar	Familiar
P5	19	Medical	Low	Not familar	Familiar
P6	19	Psychology	Moderate	Not familar	Familiar
P7	26	Computer Science	High	Familar	Not familar
P8	24	Physics	High	Not familar	Not familar
P9	19	Business Administration	Moderate	Not familar	Familiar
P10	18	Economics	Moderate	Not familar	Not familar
P11	24	Human-Computer Interaction	High	Not familar	Familiar
P12	26	Human-Computer Interaction	High	Not familar	Familiar
P13	25	Computer Science	High	Familar	Familiar
P14	30	Human-Computer Interaction	High	Not familar	Familiar
P15	25	Architecture	High	Not familar	Not familar

3.4 Experiment procedure

The experimental process follows the progressive structure of "interaction → quantitative evaluation → qualitative exploration". The internal validity of the comparative study is ensured by controlling interface variables (unified UI), balancing order effects (random grouping) and standardizing tasks. The complete experimental process includes interaction with two LLMs, completing the usability questionnaire for each model, and a short semi-structured interview. The specific framework is shown in Figure 4. Each experiment lasts 25 to 45 minutes.

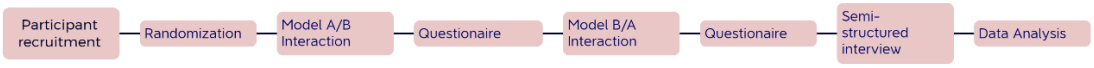


Figure 4 : Experiment procedure

A total of 15 participants were recruited for this experiment. In order to minimize learning and order effects, the order of model interaction was randomized, with seven participants testing model A first and eight participants testing model B first. Each participant completed the following three stages of experimental steps:

1. Dream interpretation task: The researcher will briefly introduce the privacy and characteristics of the model. Participants will submit personal dream descriptions to the two models respectively. The descriptions need to include the following elements: core scenes, key characters and dominant emotions, or choose a dream input model from the standardized dream case library. Participants will communicate with the model for 10-15 minutes, which can be in-depth discussions or asking about other dreams. Both models generate explanatory feedback on the content of dreams. During this period, the researcher did not directly observe the test content entered by the users, but she would observe the participants' test process and record the behavioral clues.

2. Questionnaire survey: After interacting with the two models, the participants filled out a questionnaire, which contained Likert scale questions.

3. Semi-structured interviews: Finally, the participants were interviewed in a recorded voice to collect deeper qualitative insights about their experience. The interviews focused on user preferences, perceived advantages or limitations of each model, and broader attitudes towards AI-assisted dream interpretation.

4. Data security processing: For cloud data, that is, the conversation content of chatgpt, the participants directly deleted the history. For local model data, that is, the conversation data of modelB, the researcher cleared the conversation logs in Ollama. The content of the recorded interviews is only used for this research and analysis.

3.5 Data Analysis

During the experiment, both quantitative and qualitative data were collected. The questionnaire was administered in paper format immediately after each model interaction. All interview sessions were audio-recorded and later transcribed for analysis. Quantitative data from the Likert-scale questionnaires were analyzed using paired-sample t-tests to compare users' perceptions of usability, trust, and overall experience between the two models. For the qualitative data, a thematic analysis approach was applied to the interview transcripts.

3.6 Ethical Considerations

The entire research process complies with the ethical research guidelines of Uppsala University and GDPR. Informed consent was obtained from all participants before data collection, and the purpose of the study, data security processing methods, and participants' right to withdraw at any time were clearly explained. The research process remained anonymous and tried to minimize the discomfort that participants might experience during participation. Before participants interacted with the Cloud-llm model, researcher reminded participants of the privacy risks of the interaction and deleted the records in the cloud in a timely manner after the interaction. All content that participants communicated with the AI was not stored or shared and was deleted after each experiment. For participants who were unwilling to provide personal dream information due to privacy concerns, a dream case library was provided to assist in testing. The research format was innovative to reduce participants' perception of the research relief as a test or evaluation. With written consent, researcher collected and transcribed the interview audio and video to ensure accurate recording of participants' experiences and feelings. All collected data was encrypted and stored in the way that prevented personal identification.

4 RESULT AND ANALYSIS

4.1 Qualitative analysis

4.1.1 Descriptive Statistics

This study used 15 Likert scale items in four dimensions (usability, trust and privacy, experience value, and reuse intention) to evaluate users' dangerous experience of two dream analysis models (ChatGPT vs. Private LLM). The tabular data showed that ChatGPT scored higher in the usability dimension ($M = 4.67$, $SD = 0.49$), while Private LLM scored significantly higher in the trust and privacy dimension ($M = 4.23$, $SD = 0.69$). In the other two dimensions (experience value and reuse intention), the scores of both parties were close and the difference was small. Cronbach's α of all dimensions exceeded 0.70, indicating good internal consistency and suitable for further statistical analysis.

Table 3: Paired t-tests

Dimension	Subscales	ChatGPT Mean (SD)	Private LLM Mean (SD)	Cronbach's α
Usability	Q1, Q2, Q8	4.67 (0.49)	4.20 (0.86)	0.79
Trust & Privacy	Q6, Q11-Q13	3.13 (1.15)	4.23 (0.69)	0.83
Experience Value	Q3, Q5, Q9, Q10, Q14	4.00 (0.71)	3.93 (0.80)	0.76
Reuse Intention	Q4, Q7, Q15	4.20 (0.86)	4.13 (0.92)	0.72

4.1.2Inferential Statistics

Table 4: Dimension-Level Comparisons (Paired t-tests)

Dimension	t(14)	p-value	Cohen's d	Effect Size	95% CI	Sig
Usability	3.21	.003	0.82	Large	[0.18, 0.76]	**
Trust & Privacy	-4.07	**<.001**	**1.05**	**Large**	[-1.57, -0.50]	***
Experience Value	0.38	.709	0.10	Small	[-0.31, 0.44]	ns
Reuse Intention	0.31	.758	0.08	Small	[-0.38, 0.51]	ns

4.1.3Key findings

Privacy Paradox Resolved

The Private LLM demonstrated significantly higher ratings in the Trust & Privacy dimension compared to ChatGPT ($t(14) = -4.07$, $p < .001$, Cohen's $d = 1.05$, large effect size). This finding strongly supports the hypothesis that local, on-device LLM deployment enhances users' perceived control and data security. In particular, Q11 ("I trust that no data will be leaked") showed the strongest individual effect ($t(14) = -4.24$, $p < .001$, $d = 1.09$), highlighting user concerns about confidentiality when using cloud-based AI services. These results suggest that private LLMs may effectively resolve the privacy paradox, where users desire personalized AI interactions but hesitate to disclose sensitive data due to trust concerns.

Usability Advantage of Cloud-Based Model

In contrast, ChatGPT significantly outperformed the Private LLM in the Usability dimension ($t(14) = 3.21$, $p = .003$, $d = 0.82$), reflecting a strong user preference for smoother interaction and performance. Further breakdown reveals that this difference is driven primarily by Q1 ("Ease of use") and Q8 ("Response speed"), which scored higher for ChatGPT across participants. These results indicate that while private LLMs may offer privacy benefits, they still lag behind in system-level optimization, including latency, interface fluency, and conversational polish—areas where cloud-based models backed by large-scale infrastructure currently dominate.

Equivalent Therapeutic Engagement

Despite differences in usability and trust, no significant differences were found in the Experience Value ($t(14) = 0.38$, $p = .709$) or Reuse Intention ($t(14) = 0.31$, $p = .758$) dimensions. Specifically, emotional resonance items such as Q3 ("The conversation felt meaningful") and Q5 ("I felt emotionally supported") showed similar scores across both models. This suggests that even a lightly fine-tuned private LLM can match a state-of-the-art cloud model in supporting reflective or therapeutic-like dialogue, as long as the domain is well defined (e.g., dream analysis). Thus, domain adaptation can effectively compensate for architectural limitations in generating personally resonant outputs.

Order Effect Analysis

To test whether the sequence of interaction influenced participant ratings, we conducted a between-group comparison (Group 1: model A→ model B vs. Group 2: model B→ model A). The analysis showed no significant order effect ($p = .40$), suggesting that participants' evaluation was robust regardless of the testing sequence, and supporting the internal validity of the findings.

4.1.4 Qualitative analysis

To more fully understand the impact of system design on user experience, trust, and emotional engagement, I cross-analyzed quantitative data with qualitative themes. Table 6 below summarizes the main survey indicators, significant statistical effects, corresponding qualitative themes, and user direct comments, showing the true image of the two types of LLM in the minds of users.

Table 6: Qualitative analysis user voice behind quantitative results

Dimension	Emergent theme	Illustrative quote (P = participant)	Quantitative echo
Trust & Privacy	"I own the walls"—local control lowers disclosure anxiety	"With the local bot I can just type everything. I don't worry it stays on some server." (P7)	Large advantage for Private LLM in Trust & Privacy ($d = 1.05$)
Usability	Speed & polish matter	"ChatGPT feels snappier and the wording is smoother… like a native speaker versus a draft." (P2)	ChatGPT Usability lead ($d = 0.82$) driven by ease-of-use (Q1) & latency (Q8)
Experience Value	Symbolic depth is comparable	"Both give me new angles on the snake image; the private one just takes longer." (P11)	No sig. difference in Experience Value
Reuse Intention	Conditional adoption	"If the private model gets faster, I'd keep it for my 'real' dreams and use ChatGPT for fun ones." (P9)	Reuse Intention ns; mean scores ≈4 for both

Additional cross-cutting insights:

1. **Latency tolerance ≠ disengagement**: Seven participants said they would accept slower responses if privacy were critical, aligning with the *privacy-paradox resolution* observed quantitatively.

2. **Perceived therapeutic manipulation**: Six users felt ChatGPT's constant reflective questions "prolonged the session on purpose," signalling a design tension between guided reflection and conversation length—an issue not captured by the questionaire.
3. **Cultural framing flexibility**:Four users appreciated the private model's option to choose Jungian vs. Freudian or cross-cultural frames, reinforcing the value of template-based prompt control even though this nuance was invisible in the survey.

The joint display shows complementarity: quantitative tests pinpoint where the two systems differ (privacy, usability), while qualitative evidence explains why (data-sovereignty feelings vs. latency/friction) and surfaces latent adoption strategies (hybrid use) that numbers alone cannot reveal. This integrative lens strengthens the ecological validity of my conclusions and highlights concrete optimisation targets.

5 DISCUSSION

5.1 Revisiting the research question

5.1.1 Limitation of public LLM in dream interpretation:

Users will self-censor when inputting results based on privacy considerations. Participants will deliberately modify their dream descriptions when using ChatGPT to avoid information leakage, such as blurring cheating behavior as "emotional conflict". This affects the model's recognition and interpretation of keywords to a certain extent, resulting in deviations in the generated dream interpretation. Although the cloud model scored high in emotional empathy, participants concealed content related to trauma, indicating a psychological trade-off between expressive ability and perceived safety. The Paradox of Engagement: Although follow-up prompts from the public LLM deepened reflection, they were occasionally perceived as employing addictive design (P13)—a persuasive interaction pattern that was notably absent in the private LLM, raising ethical questions about emotional manipulation in reflective AI tools. These findings are consistent with previous criticisms that cloud(LLM), while powerful, is not designed for psychologically sensitive or reflective interactions, and its privacy considerations and memory settings do not support such interactions.

5.1.2 Private LLM's Value Proposition

Due to the privacy trust of private LLM, users reveal more details when describing their dreams. Although the accuracy of symbolic interpretation of private LLM is comparable to that of cloud models, it performs poorly in terms of emotional resonance. The main reason is that the response time is too long, exceeding 4 second, which disrupts the user's reflection process. Compared with the large cloud model, the content is not deep enough and lacks active prompts. Users perceived the system as having the potential to serve as an AI-companion, helping people build personal dream information systems to better record and organize dream information without fear of information leakage. In addition, some participants expressed a sense of co-creation in the private model. Rather than using the system as a final interpretation, they used it as an aid to personal meaning-making. These interactions reflected a shift in expectations: users no longer sought a "correct" interpretation, but rather valued the model's ability to reflect, organize, and facilitate further self-reflection. Even a limited fine-tuning process helped align the model's responses with the emotional tone and symbolic structure of the user-provided data, thereby enhancing perceived relevance and interpretation accuracy.

Thus, when customized to individual use, the Private LLM showed good potential as an emotionally engaged interpretation companion, especially in an environment that prioritizes trust, reflection, and long-term engagement.

5.1.3 Users see value in adopting private LLMs in dream interpretation for personal use in the future

Participants consistently expressed interest in the idea of using a private LLM for long-term, personal dream interpretation. Several envisioned the system as a kind of digital companion for emotional tracking, symbolic journaling, or even lightweight therapeutic support. Unlike public LLMs, which felt transactional and impersonal, the private model was imagined as something that could evolve with the user—learning from recurring dream patterns, remembering symbolic motifs, and gradually building a personalized interpretative memory.

Importantly, the perceived value of such systems was closely tied to data sovereignty. Users emphasized that in order for such a tool to become part of their intimate reflective routine, it must ensure full local control, memory transparency, and customizable data deletion. This desire for a psychologically safe and persistent reflective space mirrors trends in slow tech and personal informatics, where systems are valued not for speed or scale, but for depth, continuity, and ethical design.

Overall, while current private LLMs are still technically limited in terms of fluency and symbolic repertoire, users showed strong interest in adopting them—provided they remain under user control and are designed to facilitate gradual, meaningful, and context-sensitive self-insight.

5.2 Implications for Private LLM Design

Based on the results of this study, researcher propose several design implications for future development of AI systems focused on dream interpretation. These implications aim to balance technical capability with ethical alignment and psychological sensitivity, particularly in systems that engage with users' private narratives and affective content.

1. **Prioritize Data Privacy and User Control**
   Users showed significantly higher trust and willingness to share personal content with local models. This suggests a need for explicit design cues that highlight data sovereignty and local processing. Systems should offer transparent privacy policies, reinforce that no data is transmitted to the cloud, and include UI signals (such as "running locally" badges or privacy toggles) that reinforce a sense of control.
2. **Support Personalization and Long-term Engagement**
   Participants envisioned long-term use of a private interpreter as a personal journaling assistant. Designers could explore features such as dream memory systems that track recurring symbols, emotions, or themes, and evolve over time. Enabling users to build a symbolic lexicon or maintain a reflection log could enhance engagement and foster psychological insight.
3. **Embrace Reflective, Dialogic Interpretation Styles**
   Rather than offering deterministic interpretations, systems should adopt suggestive and reflective language. Participants were more comfortable when interpretations were framed as possibilities rather than facts. Language strategies that encourage self-reflection (e.g., "You might be experiencing...") could help users make meaning while maintaining agency.
4. **Enable Multiple Interpretive Perspectives**
   Users expressed interest in choosing between different symbolic frameworks or narrative styles (e.g., Freudian, Jungian, mythological, cultural). Designing for interpretive plurality not only supports personalization but also fosters user exploration. This could be implemented through customizable "interpretation modes" or transparent model explanations.

These design implications highlight the need to treat dream interpretation systems not merely as technical tools, but as emotionally sensitive, trust-driven, and user-guided interfaces. Future designs should aim for systems that empower users to explore their inner world safely, meaningfully, and on their own terms.

6 CONCLUSION

This study explored the design potential of a private, locally deployed LLM as an intelligent companion for dream interpretation. By combining personal data sovereignty, symbolic analysis, and affective computing, Researcher conceptualized a system that enables users to engage with their dreams in a secure, introspective, and meaningful way. Through literature review, prototyping, and user studies, researcher examined the comparative experiences of two AI models: a cloud-based LLM and a locally deployed private LLM. Findings indicate that while users appreciated the interpretive richness of cloud-based models, they expressed significantly greater trust and willingness to share sensitive information with the private system—highlighting the importance of privacy, personalization, and emotional resonance in affective AI applications.

These results suggest that private LLMs may be better suited for emotionally sensitive domains like dream journaling and self-reflection, offering users not only data ownership but also a psychologically safe space. Moreover, user feedback reveals an appetite for long-term engagement, symbolic memory tracking, and reflective dialogue—pointing toward future design directions that go beyond one-time AI responses.

Ultimately, this research contributes to the emerging discourse on psychologically aware and privacy-respecting AI systems, and suggests new opportunities for designing transparent technological partners" that support personal meaning-making in intimate human contexts.

6.1 Limitation:

This study has several limitations that should be considered when interpreting the findings. With only 15 participants, the statistical power of the results is limited. A larger and more diverse sample would be needed to generalize the findings to a broader population. Most participants were students or individuals with prior experience using AI systems. This may have influenced their expectations and ability to critically evaluate the models. Although both models were tested with the same questionnaire and under similar conditions, the order of exposure may still have influenced participants' ratings, despite attempts to counterbalance the sequence. Both dream content and AI responses are highly subjective. Although thematic analysis helped capture qualitative insights, personal biases—both from participants and researcher—might affect interpretation.

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A APPENDICES

A.1 Appendix A: Pre-Study Interview Guide – Understanding Users' Dream Interpretation Practices and Expectations

This semi-structured interview guide was used in the early stage of the study to understand participants' existing approaches to dream interpretation, their emotional needs, and their expectations for an AI-powered, locally deployed dream interpretation assistant.

Background & Current Practice

1. Can you tell me about your usual approach to understanding or reflecting on your dreams, if any?
2. Have you ever shared your dreams with others (e.g., friends, therapists, or online forums)? What motivates or prevents you from doing so?
3. Do you currently use any digital tools (apps, websites, AI) to help interpret your dreams? If yes, which ones and what is your experience with them?

Perceived Role of Dream Interpretation

4. What role do you think dream reflection plays in your emotional or psychological life, if any?
5. Have you ever had a dream that you felt was too personal or sensitive to talk about? How would you prefer to process such dreams?

Expectations for a Private LLM Tool

6. If you had access to an AI that could help you interpret your dreams, what kind of support or insights would you hope to receive?
7. What emotional qualities would you expect this AI tool to have (e.g., empathetic, neutral, honest)? Why are those important?
8. Would you feel more comfortable using such a tool if it was fully local and did not store your data in the cloud? Why or why not?
9. In what ways do you think a private dream interpretation assistant could be helpful in your day-to-day life or long-term self-reflection?
10. Is there anything you would not want such a system to do or assume? What boundaries would be important for you

Dream Interpretation AI Evaluation – Likert Scale Questionnaire (1–5 points)

Please rate each statement below based on your experience using the model.
Scoring guide:
1 – Strongly Disagree | 2 – Disagree | 3 – Neutral | 4 – Agree | 5 – Strongly Agree

The model you used: _____

No.	Statement	Score (1–5)
U1	This AI is easy to use	1 2 3 4 5
U2	I can understand the AI's dream interpretation	1 2 3 4 5
U3	The AI's response made me feel it understood my dream	1 2 3 4 5
U4	I would reuse this AI for dream interpretation	1 2 3 4 5
U5	The feedback made me feel dream interpretation is meaningful	1 2 3 4 5
U6	I am willing to share private dream content with this AI	1 2 3 4 5
U7	I believe this AI interprets dreams objectively	1 2 3 4 5
U8	I am satisfied with the AI's response speed	1 2 3 4 5
U9	Overall, the dream analysis experience is pleasant	1 2 3 4 5
U10	This AI can gradually understand my needs	1 2 3 4 5
U11	I trust that this model won't leak my dream content	1 2 3 4 5
U12	I feel safe and worry-free when inputting dream content	1 2 3 4 5
U13	I understand and accept how this model handles data (local/cloud storage)	1 2 3 4 5
U14	I think this model is suitable for self-exploration and emotional support	1 2 3 4 5
U15	If continuously improved, I would continue using this AI in the future	1 2 3 4 5

A.2

A.3 Questionaire

Semi-structured Interview

Understanding and attitude of dream interpretation

1. Are you familiar with dream interpretation? Have you read related books or theories?
2. What do you think the dream expresses? Is it your private information?
3. Have you ever consulted someone for dream interpretation or fortune telling? Did you trust the results?

Comparing the two models

4. Which model's interpretation did you prefer? Why? (accuracy, personalization, trustworthiness?)
5. Did you test the models with the same dream multiple times? How did the responses differ?
6. Do you see future applications for AI-based dream interpretation? In what context?

AI-related trust &privacy

7. Would you share personal information with AI? Why or why not?
8. Are you concerned about AI leaking your personal data? Would this affect your experience?
9. Will you change the questions you enter because AI may steal your information? Does this affect your experience?
10. If you know that the LLM in your computer will not leak your information and has growth potential. Will you change the way you communicate with AI now?